

electricity meter would show a saving of about 2.1 per cent and when you have only light and fan load, your electricity meter would show a loss of about 1.8 per cent (although PF has improved a bit, more power has been delivered to the load). Here are the calculations.

Case I: $(2.341 - 2.292) / 2.341 \times 100 \approx 2.1\%$ (Lower consumption shows saving)

Case II: $(254.8 - 258.2) / 254.8 \times 100 \approx -1.8\%$ (Higher consumption indicates loss)

Energy Star rating of home power savers

Given below is what ENERGY STAR (a joint program of the US Environmental Protection Agency and the US Department of Energy) has to say about these home power savers:

"Power factor correction devices claim to reduce residential energy bills and to prolong the productive life cycles of motors and appliances by reducing the reactive power (kVAR) that is needed from the electric utility.

"We have not seen any data that proves these types of products for residential use accomplish what they claim. Power factor correction devices improve power quality but do not generally improve energy efficiency (meaning they won't reduce your energy bill). There are several reasons why their energy efficiency claims could be exaggerated. First, residential customers are not charged for kVA-hour usage but by kilowatt-hour usage. This means that any saving in energy demand will not directly result in lowering a residential user's utility bill. Second, the only potential for real power savings would occur if the product were put in the circuit only while a reactive load (such as a motor) were running, and taken out of the circuit when the motor is not running. This is impractical, given that there are several motors in a typical home that can turn on at any time (refrigerator, air-conditioner, water pump, HVAC blower, vacuum

		ELECTRONICS REGIONAL TEST LABORATORY (NORTH)			
TEST REPORT					
TEST REPORT NUMBER			DATE		PAGE NO.
ERTL/NY90(4)-3K7070UC0357/SYS			20/07/2007		01 OF 03
Service Request No.0770015			Date : 10/07/2007		
Name & Address of the Client			B.S Energy, Khasra - 883,Nala Road, Village- Rihala , Rohini.		
Description/ Identification of the Item: -					
Nomenclature			IDEAL POWER SAVER		
Model No./Type No.			IP5010		

SL.No.	CL No.	Parameter	Nominal Value/ Requirements	Measured Observations	Value/
1.0	-	2 Tubes + Air Conditioner + Refrigerator. at 225.4V, 49.41Hz. Apparent Power	Not specified	Without Energy Saver 2.611kVA	With Energy Saver 2.495kVA
		Current		11.584Amp.	11.114Amp.
		Power factor		0.895	0.918
		Real Power		2.341kW	2.292kW
				At 224.5V & 49.57Hz.	
2.0	-	3 Tubes + 2 Fan At 231.3V, 49.69Hz. Apparent Power	Not specified	Without Energy Saver 269.1VA	With Energy Saver 265.1VA
		Current		1.1637Amp.	1.1447Amp.
		Power factor		0.947	0.974
		Real Power		254.8W	258.2W
				At 231.6V & 49.65Hz.	

Fig. 3: A test report

cleaner, etc), but the unit itself is intended for permanent, unattended connection near the house breaker (circuit breaker) panel.

"For commercial facilities, power factor correction will rarely be cost-effective based on energy savings alone. The bulk of cost savings power factor correction can offer is in the form of avoided utility charges for low power factor. Energy savings are usually below 1 per cent and always below 3 per cent of load, the higher percentage occurring where motors are a large fraction of the overall load of a facility. Energy savings alone do not make an installation cost-effective."

Comments of a power saver distributor

In a report published online at <http://electricitysaver.com.au/home/index.php/how-does-it-work>, one of the erstwhile distributors of A2-Power

Saver-250 says:

"We were once on distributors of a power correction based electricity saving device, what we called the 'electricity saver' - a product that claimed 30 per cent cost savings. After rigorous testing and third-party independent tests, we decided to discontinue distribution of this product due to its false claims. Similar products are found all over the Internet, all with the same false claims. Many are still available. To help educate the public, we've kept this page online.

"So do these devices work? Simply put, no.

"Products

like these have been independently tested and found to not save you money or reduce your electricity consumption. Since posting this, we've received many threats from dubious retailers and distributors claiming of anecdotal evidence of savings. But all have declined to send a product for some real lab testing.

"We've also heard reports of new products hitting the market under various names and packaging with similar claims, with some going as far as filming fake testing, or including test reports which actually don't confirm any power savings."

This item is even now advertised at alibaba.com.

Are all these power savers a hoax?

Not exactly, but most of the power saver manufacturers do misguide or exaggerate the advantages of their